Ex.No.4: OFDMA Resource Block Allocation

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access)resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Objective

1. To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.
2.To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.
3.To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;
% System parameters
bandwidth = 10e6;
rb_size = 180e3;
% Total Resource Blocks
% 10 MHz
% 180 kHz per RB
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users)))
+ 1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
disp('RB Allocation per User:')
disp(alloc)
figure
% -------- Graph 1: RB Allocation per User --------
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: RB Utilization --------
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
```

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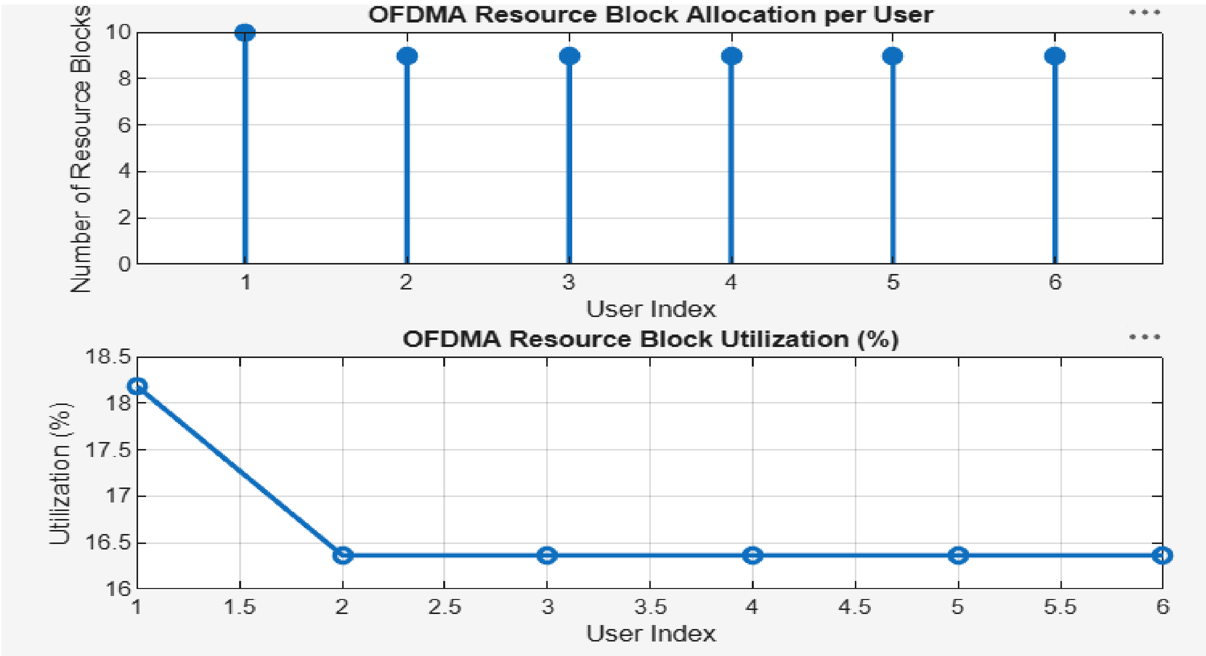
```
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```



Objective 2 Matlab Code and Output:

```
clc; clear; close all;
% Total system resource blocks
total_RB = 50; % Number of active users
users = 1:5; % RB allocation to users (example scenario)
alloc = [12 10 9 8 11]; Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util')
figure
% -------- Graph 1: RB Allocation --------
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
```

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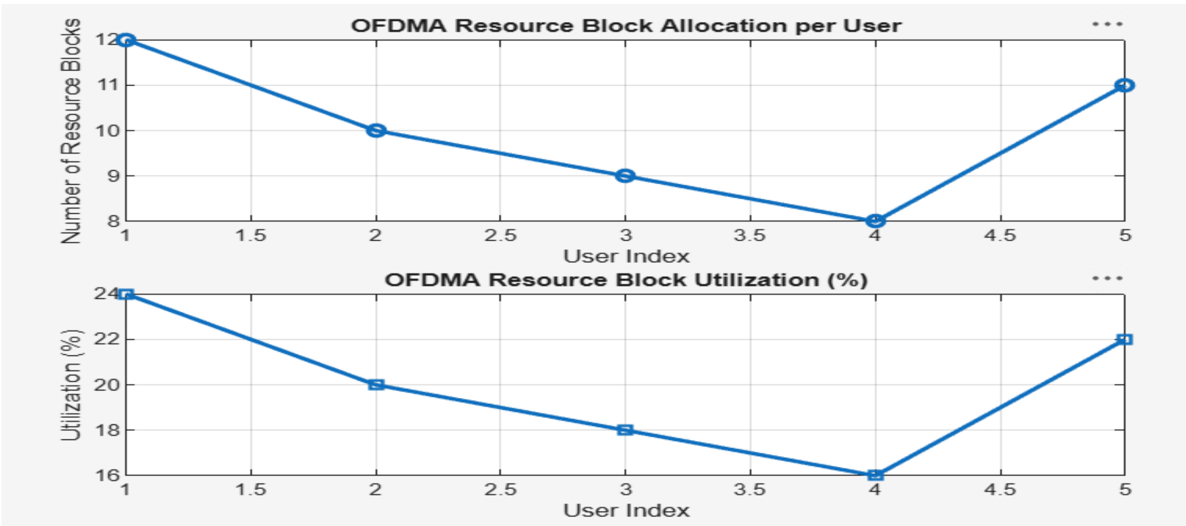
```
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```



Objective 3 Matlab Code and Output:

```
rb_bw = 180e3;
users = 1:5;
% Bandwidth per RB (180 kHz)
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2];  % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')
figure
% -------- Graph 1: Throughput per User --------
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
```

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```
% -------- Graph 2: RB vs Throughput --------
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```